

Analysis of groundwater quality using water quality index: A case study of Simhadripuram Mandal, YSR Kadapa District, Andhra Pradesh.

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Abstract - Greater than 50% of the population depends on groundwater for drinking water. Groundwater is also one of our most important sources of water for irrigation. Regrettably, groundwater is at risk of pollutants. Groundwater contamination occurs when anthropogenic products such as gasoline, oil, road salts, and chemicals get into the groundwater so, it is reasonable to be converted into unsafe and unfit for human beings to make use of. The resources from the earth's surface are able to move all the way through the soil and closing stages in the groundwater such as fertilizers and pesticides are capable of discovering their approach into the groundwater supplies in excess of time. Road salt, toxic chemicals from mining sites, and used motor oil also may leak into the groundwater. To check whether the water quality is in compliance with the standards, and hence, suitable or not for the designated use. It is to know details about various Physico-chemical parameters such as Hydrogen Ion Concentration (pH), Electrical Conductivity, Total Dissolved Solids (TDS), Alkalinity, Chloride, Total Hardness, Calcium, Sodium, Potassium, and Magnesium. Groundwater is suitable for drinking purposes used but continuously long period contaminated the groundwater may be unsuitable for humans.

Key Words: Physico-chemical parameters, Toxic chemicals, Groundwater.

1. INTRODUCTION

Water quality is the determination of the appropriateness of water for a particular user depends on specific physical, chemical, and biological characteristics. Estimation of the quality of the groundwater, and these characteristics are the collected water is acceptable or not for drinking, agriculture, industries, and other a few applications (Aghazadeh et al., 2010; Doneen et al., 1964). Determining water quality necessarily the measurement and analysis of specific characteristics which include such parameters, pH, Total Hardness, TDS, and conductivity to determine if the water is suitable for a particular use. Chemical parameters are analyzed in a laboratory from samples collected in the water body of interest. The decreasing water of quality is dangerous for humans, animals, and plants. Waterborne illness and chemical concentration end up spreading among humans. Poor quality of water may be harmful to any industry related to its utilization.

Objectives

- The different thematic maps are using by the satellite data along with accuracy data such as maps of Geology, Geomorphology, Drainage, and Lineament.
- The chemical parameters are measured by different instruments with different methods in the laboratory.
- Finally analyzed data represented as graphs.

2. Materials and Methods

2.1 Study Area

The study region is located in the Survey of India (SOI) Toposheet No: 57 J/02 in 1:50,000 scale and lies between $78^{\circ}6'0''\text{E}$ to $78^{\circ}12'0''\text{E}$ and $14^{\circ}30'0''\text{N}$ to $14^{\circ}36'0''\text{N}$ with an aerial extent of 294,177,357 sq.m. The area is a semi-arid climate. The annual average rainfall 89.5mm and the temperature is 40°C . The location map of the study region is shown in Fig (1).

2.2 Geology

The Archean group is underlined by the Cuddapah basin. It is consisting of Peninsular Gneissic Complex Granite Gneisses, Schist, and Granitoids with acidic and basic intrusive. The study area mainly falling the Tadipatri formation of the Chitravati group in the Cuddapah supergroup. The main lithological unit consists of chiefly Tadipatri Formation of Dolomite, Quartzite, Volcanic Flows, and Shale.

The Geomorphology of the study area predominantly consists of the structural and Denudational hills and valleys presented due to Mogamureru River and weathering and erosion of surrounding lithological units. The remaining small amount of anthropogenic landform is also presented due to human activities. Study region geology and geomorphology maps are shown in fig (2)

The study area in joints, lineaments, and faults is presented. The groundwater recharged through these structures, so these are all significant for the groundwater prospect. The area of study drainage and lineament maps is shown in fig (3).

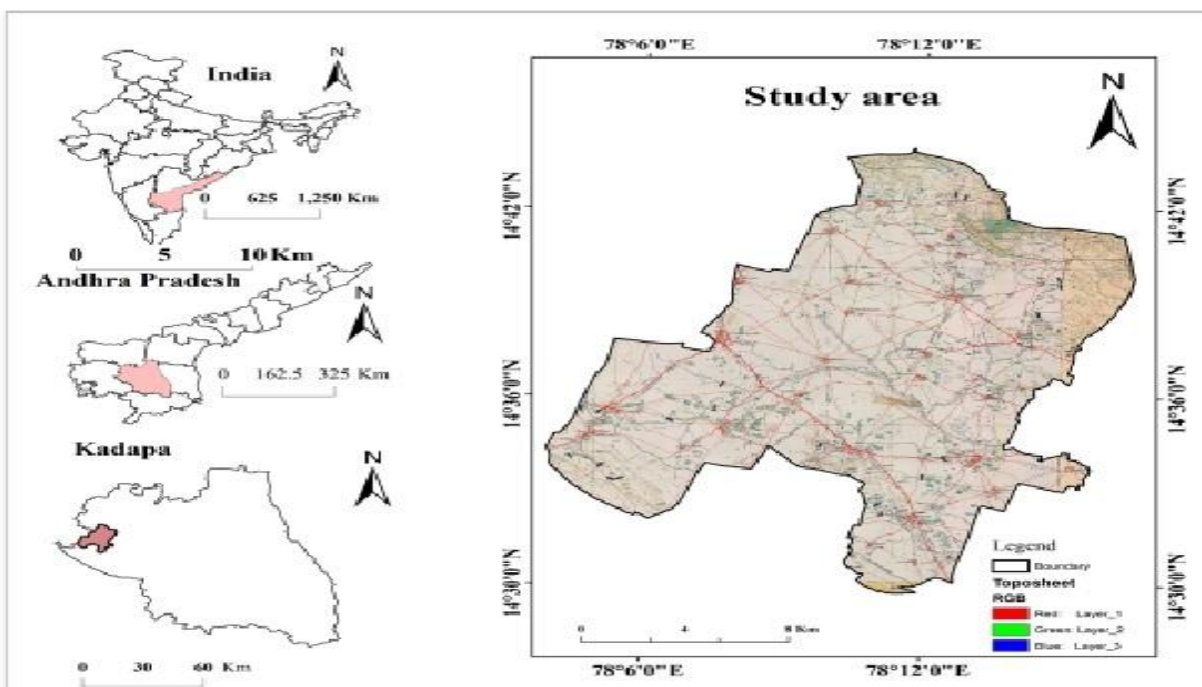


Figure 1: Location of the area of study

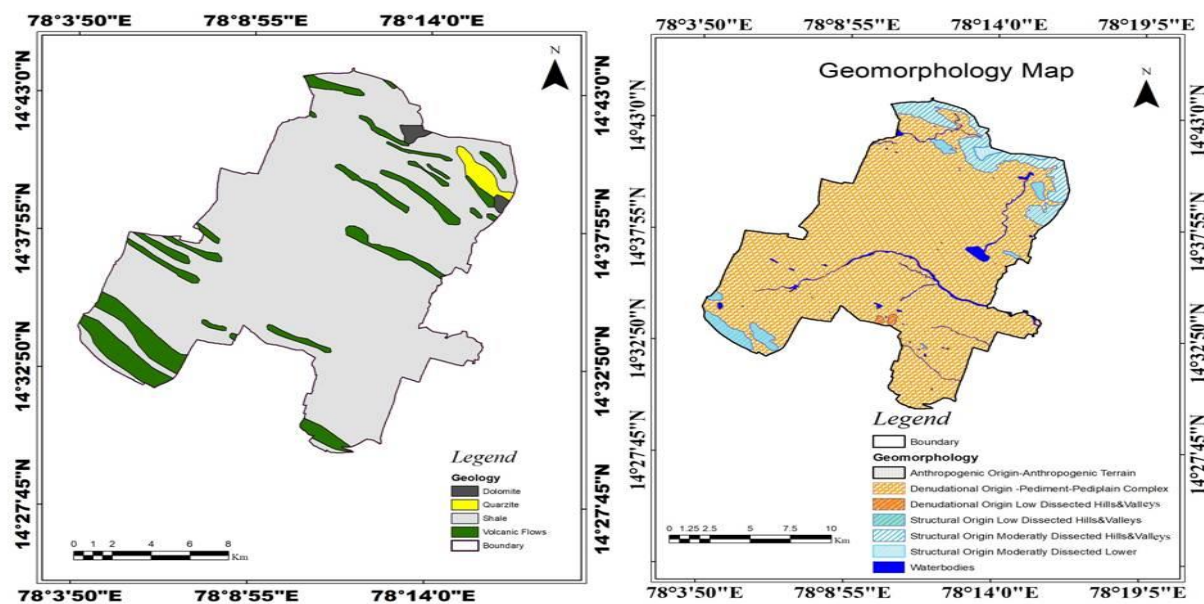


Figure 2: Geology and Geomorphology map of the area of study

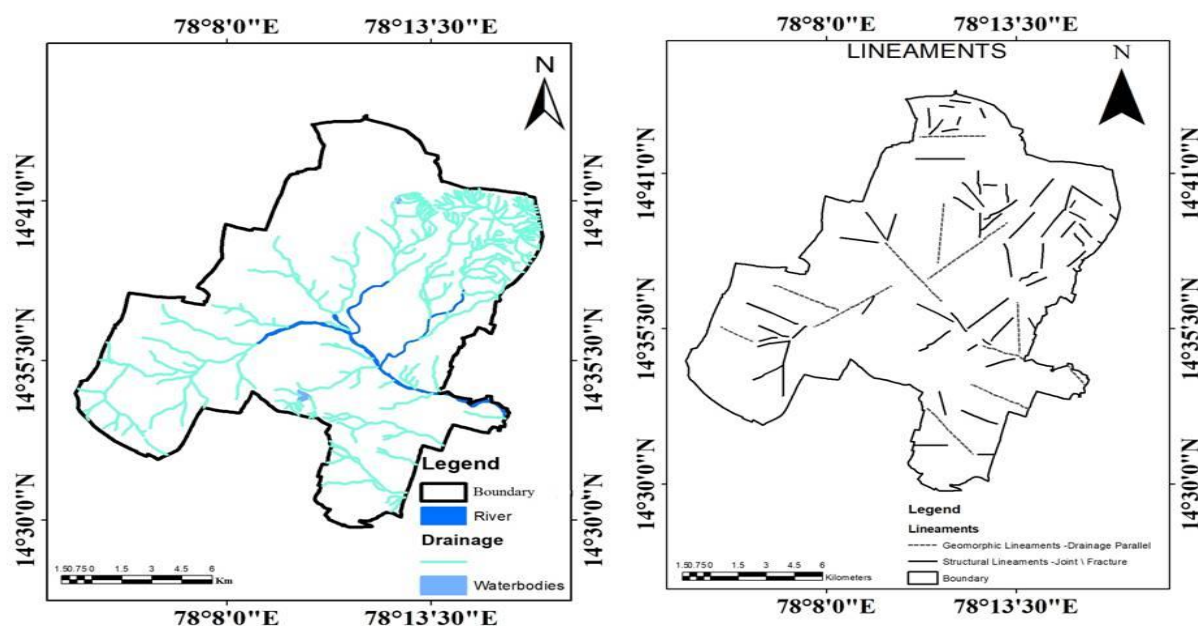


Figure 3: Drainage and Lineament map of the area of study

3. METHODOLOGY

Twenty - two samples are collected from the different villages of Simhadripuram Mandal of YSR District. The samples are collected in the month of January 2019. Pre cleaned sterilized polyethylene of one - liter capacity of the bottles taken and collected the water samples. After that samples are analyzed with different chemical parameters by using different instruments and methods. The methodology of the study area is shown in the flow chart (Fig 4).

4. Result and Discussion

4.1 Hydro geochemical characteristics

The groundwater samples are analyzed for post-monsoon to determine the relationship between the groundwater of the chemical properties with hydrogeology in the study region. The groundwater samples of hydrochemical analysis of results are represented in Table 1 and also in graphical representation.

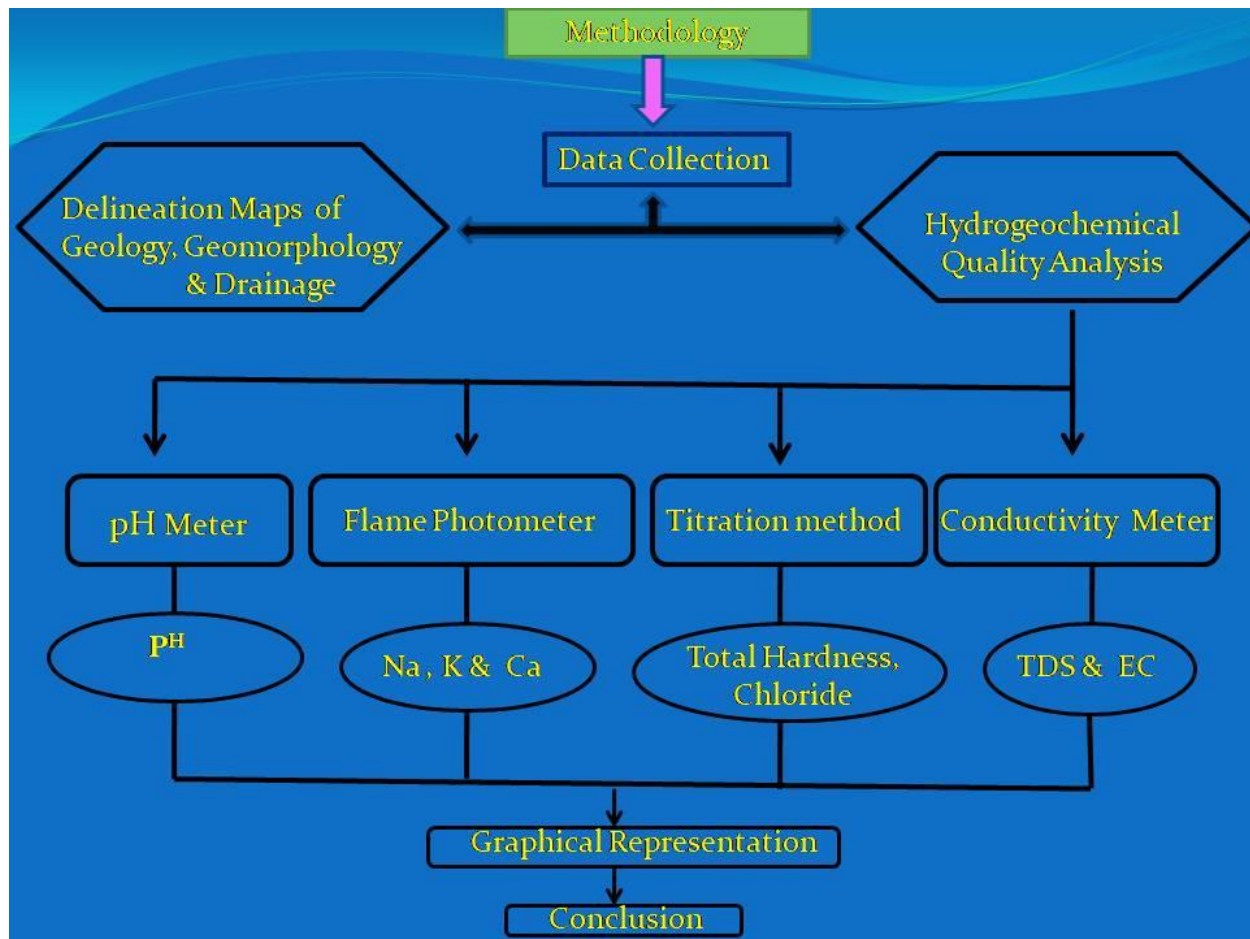


Figure 4: Methodology of the area of study

4.2 Physical Parameters

4.2.1 Hydrogen Ion Concentration (pH)

The pH parameter is significant in determining the alkalinity or acidity nature of the groundwater. pH is positively related to the electrical conductivity and total alkalinity of the groundwater (Das et al., 2015; Freeze et al., 1979). pH is mostly controlled by the precipitation of the different minerals and also controlled by ionic concentrations of carbon dioxide (CO_2), carbonates (CO_3^{2-}) and bicarbonates (HCO_3^{2-}) (Hem 1991). The acceptable confined value is 6.5 to 8.5 pH for the purpose of drinking according to BIS (2012). The pH range from 7.5 to 8.8 in the area of study. The groundwater samples are falling the slight alkalinity nature. Graphical representations of pH in the study area shown in Figure 5.

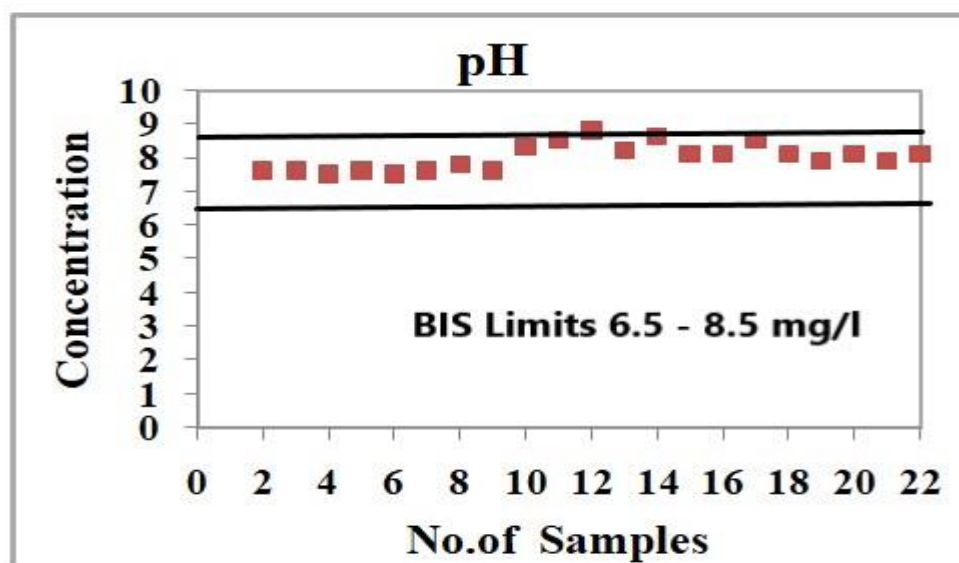


Figure 5: Graphical representations of the pH in the area of study

4.2.2 Total Hardness

Hardness is due to ions in cations of calcium, magnesium, and alkaline earth metals. The permanent hardness comes from the elements of predominantly sulfates (SO_4^{2-}) and chlorides (Cl^-) are present in the groundwater. The temporary hardness comes from the elements of predominantly calcium (Ca^{2+}) and magnesium (Mg^{2+}). The hardness is a relationship directly proportional between the rate of filtration and dissolution. The geological formations are mainly the reason for the calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions in the study area (Gibbs et al., 1970; Hemet al. 1980;Islam et al., 2015;Ibraheem et al.,2017). The degree of hardness in water is commonly based on the following classification

Table 2 Hardness classification of water

S.No	Hardness mg/l CaCO_3 as	Water Class
1	0-60	Soft
2	60 -120	Moderately Hard
3	120 -180	Hard
4	>180	Very hard

The permissible confined value is 200 mg/l to 600 mg/l (BSI, 2012) total hardness for drinking purposes. Hardness range from 37 to 4730 mg/l in the area of study. The interactions of water with the geological formations are the main reason for the hardness of the study area. Granitic rocks are significant for the hardness of the groundwater. The entire study region's hardness overpasses the desirable confined value. Graphical representations of TH in the study area shown in Figure 6.

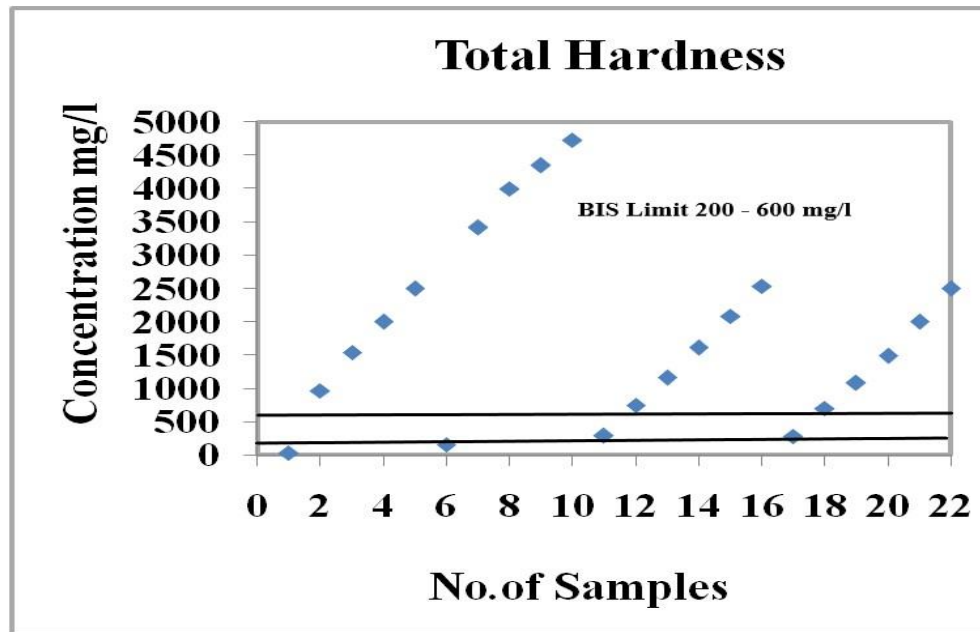


Figure 6: Graphical representations of the Total Hardness in the area of study

Table 1: Hydrogeochemical Parameters of groundwater of the area of Study

S.No.	Total Hardness	Cl	CO ₃	HCO ₃	pH	EC	TDS	Ca	Mg	Na	K
Units	Mg/L	Mg/L	Mg/L	Mg/L		us/cm	ppm	Mg/L	Mg/L	Mg/L	Mg/L
1	37	50	132	488	7.6	1038	623	8.8	90	20	16
2	970	50	210	610	7.6	1186	724	2.7	135	3	5
3	1540	291	36	207	7.5	1020	633	9	48	10	17
4	2000	142	150	427	7.6	1306	810	8.5	151	0	16
5	2500	169	112	817	7.5	1001	621	9.1	59	20	17
6	160	117	144	1610	7.6	923	582	8.4	35	90	13
7	3420	557	-	305	7.8	1715	1081	7.3	153	30	10
8	4000	251	87	226	7.6	1142	720	8.6	96	20	17
9	4350	82	-	250	8.3	754	476	9.5	86	10	16
10	4730	220	51	177	8.5	907	581	8.1	77	10	13
11	290	185	30	189	8.8	850	544	8.5	39	60	16
12	750	170	-	281	8.2	981	638	8.9	123	40	7
13	1160	398	-	220	8.6	1187	772	7.3	117	0	13
14	1610	209	-	323	8.1	1046	680	3.8	40	60	16
15	2090	142	33	305	8.1	987	682	8.9	96	30	17
16	2540	117	39	177	8.5	825	545	8.1	136	40	15
17	280	121	33	122	8.1	600	396	7.6	61	30	13
18	700	92	63	256	7.9	813	537	8.8	64	20	16
19	1080	114	33	250	8.1	868	582	9.1	50	90	12
20	1490	209	45	165	7.9	722	484	8.4	26	10	14
21	2000	614	-	216	8.1	1470	985	8.6	122	0	14
22	2500	114	-	189	7.9	665	446	9.2	70	40	14

4.2.3 Electrical conductivity

Electrical conductivity is a crucial role in the salinity parameter measurement of the groundwater. The permissible standard for salinity is 900 $\mu\text{S}/\text{cm}$ according to the State Water Source, Control board for drinking purpose (Nagaraju et al., 1972). The study area of Electrical conductivity (EC) value between 600 $\mu\text{S}/\text{cm}$ -1715 $\mu\text{S}/\text{cm}$ at 25°C presented. Based on the electrical conductivity concentration the groundwater can be classified as six classes according to the Rhoades et al and this is shown in Table 2.

Table 3 Classification of Electrical conductivity concentration

Type	EC Value in $\mu\text{S}/\text{cm}$	Types of Water
1	< 700	Non- Saline
2	700 -2000	Slightly Saline
3	2000	Moderately Saline
4	2000 -10,000	Highly Saline
5	10,000 -25,000	Very high Saline
6	25,000 – 45,000	Braine Saline

4.2.4 Total dissolved solids

Total dissolved solids (TDS) are significant parameter quality of the groundwater. Total dissolved solids (TDS) are commonly known as the concentration of all dissolved minerals in the groundwater. TDS is a high concentration presented in the groundwater gastrointestinal irritation produced. The permissible limit of total dissolved solids (TDS) is 500 mg/l to 2000 mg/l (BIS, 2012), and TDS range from 396 mg/l to 1081 mg/l in the study area. TDS level above 1000 mg/l is poor tasting and considered as suitable for drinking purposes. TDS level above >2000 mg/l is considered as suitable for drinking purposes. Graphical representations of TDS in the study area shown in Figure 7.

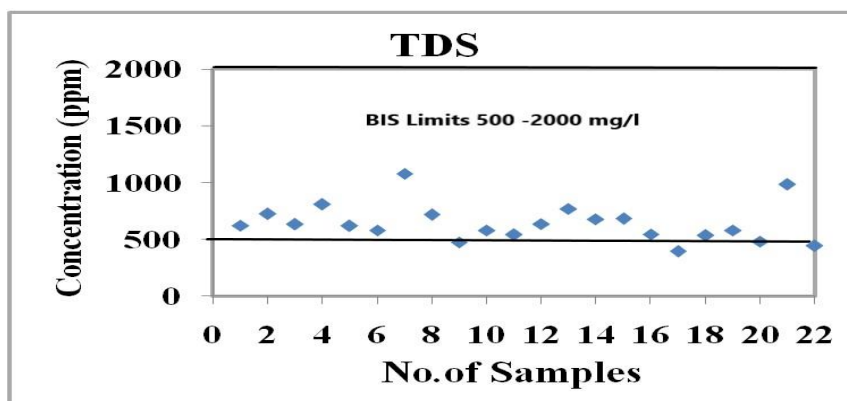


Figure 7: Graphical representations of the TDS in the study area

4.2.5 Sodium

All groundwater consists of some sodium because of surrounding lithological units and soils were as easily dissolved sodium compounds. Erosion of sodium bearing rocks, precipitation, irrigation, and sewage. The permissible confined value is 200 mg/L (WHO 2012) for drinking water purposes (Raghunath et al., 1987; Singaraja et al., 2014). The sodium concentration of the area study is 0 to 90 mg/l. A higher concentration of sodium may be problematic for peoples with hypertension, heart disease or kidney problems

4.2.6 Potassium

The potassium concentration (K^+) is lower than the concentration of sodium due to the slow rating of weathering of the potassium bearing rocks than those of sodium bearing rocks. The potassium comes from predominantly the decomposition of the clay mineral in silicates. The potassium can also add in the groundwater through the pesticides and fertilizers. A higher concentration of potassium indicates a laxative effect. It causes kidney diseases or other problems such as heart disease, hypertension, diabetes, and coronary artery diseases. The allowable limit of potassium is 1–8 mg/L (WHO 2011). The potassium concentration of the area of study is 5 to 17 mg/l presented.

4.2.7 Calcium

The calcium ion is the main source of the geological formation of the study area. The geological formation especially calcareous rocks in limestone and dolomite. Low concentration of calcium are the risk of hypertension, nephrolithiasis, coronary artery, colorectal cancer, and insulin resistance diseases increased. The higher concentration of calcium in the groundwater very use full for bladed stones or kidney stones problems are avoided. The confined value of calcium (Ca^{2+}) concentration is 75 mg/l to 200 mg/l (BIS, 2012) for drinking purposes. The calcium concentration range from 2.7 mg/l to 9.5 mg/l in the area of study. Graphical representations of calcium in the study area shown in Figure 8.

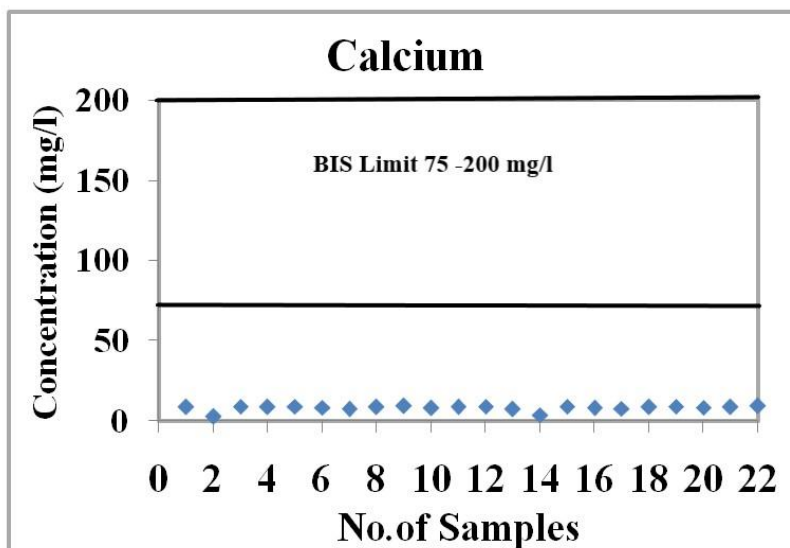


Figure 8: Graphical representations of the calcium in the area of study

4.2.8 Magnesium

The magnesium is the main cause of certain rocks and minerals especially limestone and dolomite whereas dissolved these rocks and minerals in the groundwater. The high concentration of magnesium in the groundwater circulatory or heart problems, eclampsia in pregnant women, hypertension, produced in humans (Paliwal at al., 1972). The acceptable limit of magnesium range from 30 mg/l to 100 mg/ l (BIS, 2012) in the study area. The magnesium range from 26 mg/l to 196 mg/l (BIS, 2012) in the area study. Graphical representations of magnesium in the study region shown in Figure 9.

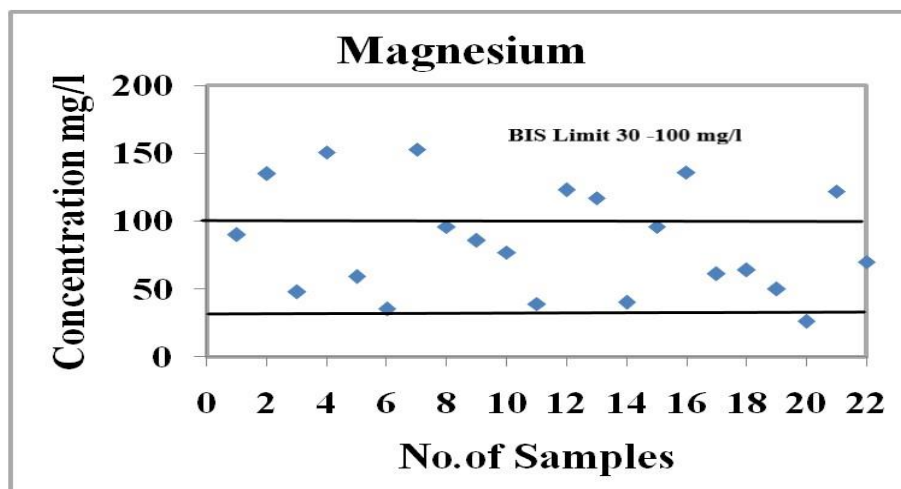


Figure 9: Graphical representations of the magnesium in the area of study

4.2.9 Chloride

Chloride in groundwater found caused by the process of chloride leaching, which has chloride contains related to the rocks and comes from anthropogenic activities such as chemical industries, seaware, and disposal. The permissible limit of chloride 250 mg/l to 1000 mg/l (BIS, 2012) in the study area. The chloride range from 49.7 to 614.1 mg/ l in the area of study. Graphical representations of chloride in the study region shown in Figure 10.

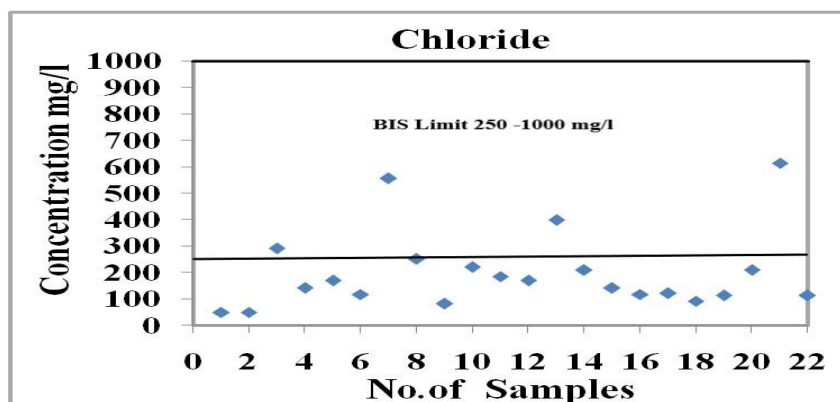


Figure 10: Graphical representations of the chloride in the area of study

4.2.10 Total Alkalinity (CO_3 and HCO_3)

Carbonates and Bicarbonates are highly present in the groundwater due to the reaction between water, calcium, and magnesium in the form of a solution in the soil. Finally precipitated in the form of calcite and magnesite minerals (Standard et al., 2014).

The permissible limit of carbonate (CO_3) in drinking water is 10 mg/l and the rejection limit is 50 mg/l. The carbonate concentration range from 0 to 210 mg/l in the study area. The acceptable limit of bicarbonate (HCO_3) 500 mg/l in the study area. (Todd, 1980). The bicarbonate concentration of the groundwater in the study area is ranging from 122 mg/l to 1610 mg/l and Most of the water samples of the study area contain no carbonate ions.

5. Conclusion

The study region of the assessment of the groundwater sample for purposes of drinking and irrigation depends upon the Indian standard ranges and it reveals that the P^{H} is within the desirable limit. In the study area, as falls ionic concentration order $\text{HCO}_3 > \text{Cl} > \text{CO}_3 > \text{Mg} > \text{K} > \text{Ca}$ in the groundwater. The Calcium, Chloride, and TDS are within the desirable limit hence the ground water utilized for drinking and irrigation. The EC, Sodium, Potassium, Magnesium Bicarbonates, Carbonates, and Total Hardness is slightly exceeding ranges from the Indian standards (Table 4). So, the salinity hazard groundwater is also suitable for purposes of drinking used but long years to years may be started health problems.

The pH range from 7.5 to 8.8 in the area of study, so it is representative of the alkaline nature of the groundwater. The pH values are exceeded desirable limits. The electrical conductivity range from 1060 $\mu\text{Siemens/cm}$ to $\mu\text{Siemens/cm}$ at 25°C . The Total Hardness range from 37 to 4730 mg/l in the study area. The Total Hardness is a Groundwater of the entire study area that exceeds the desirable limits. The calcium range from 2.7 mg/l to 9.5 mg/l in the area of study.

The bicarbonate range from 122 mg/l to 1610 mg/l in the area of study. The bicarbonate (HCO_3^-) ionic concentration is a dominant anionic concentration in natural water. Dissolution of calcite and dolomites are the main reason for the bicarbonates are present in the groundwater. The permissible confined value of chloride concentration for drinking water is 49.7 mg/l (ISI, 1983). The chloride (Cl) ranges from 50 to 614 mg/l in the area of study. The decomposition of sedimentary rocks and evaporates are reasons for chloride in the groundwater.

Table 4: Physico - Chemical Parameters of groundwater of the area of study

Chemical Parameters	Indian Standards	Study area Values
pH	6.5-8.5	7.5 - 8.8.
TDS(Total Dissolved Solids)	500-2,000	396-1081
Chloride	250-1,000	50-614
Total Hardness	300-600	37-4730
Calcium	75-200	2.7-9.5
Bicarbonates	244-732	122-1610
Magnesium	30-100	26-153

The chloride high concentrations in the groundwater because of salt beds are presented and also fertilizers, waste of animal, plant, and domestic into the groundwater. The groundwater in magnesium ion main source due to the dolomite, evaporate, magmatic rock minerals. The sodium (Na^+) ion is presented due to the decomposition of predominantly plagioclase minerals, the exchange of clay, and degradation of evaporates (NaCl). The potassium (K^+) coming to the groundwater due to the decomposition and degradation of the minerals of feldspars in predominantly orthoclase and microcline and clay.

ACKNOWLEDGEMENTS

The first author C. Krupavathi, greatly thankful to Department of Science and Technology (DST), Government of India, for financial support through Inspire programme (Sanction order No. DST/INSPIRE **Fellowship/2017/IF170787**). Also thankful to, Department of Geology, Yogi Vemana University, for necessary facilities for carrying out my research work.

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